

The Role of AI-Mediated Feedback in Shaping Attention, Motivation, and Media Engagement

Songlin Xu

Who Controls Your Attention?

Power being restored after disruption caused by Storm Chandra

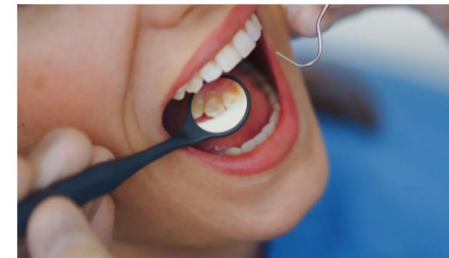
Warnings for heavy rain, flooding and strong winds have been issued across Northern Ireland.

3 hrs ago | Northern Ireland



Man, 19, charged with arson after house fire

The fire happened in Crannog Way, Dungannon. No one was in the house or injured.



'My tooth gap feels like a goalpost - but I can't get an NHS dentist'

Joe Friel has tried and failed over a four-month period to find a dentist to treat him under the NHS.



Listen: North West Today

A round-up of local news, sport and weather across the north west of Northern Ireland.

(Figure source: [bbc.com](https://www.bbc.com))

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Not just content,
but **feedback** shapes attention.

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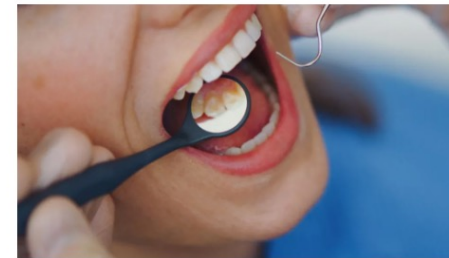
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Highlighted Article

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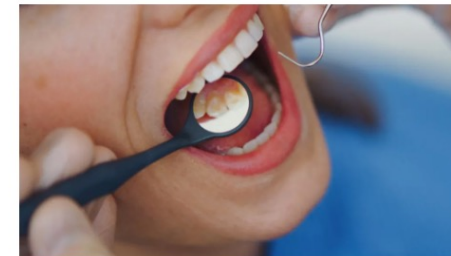
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Ranked Articles



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Feedback Modalities

- a. Highlights
- b. Rankings
- c. etc.

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Today's class:

Experience → Learn → Apply

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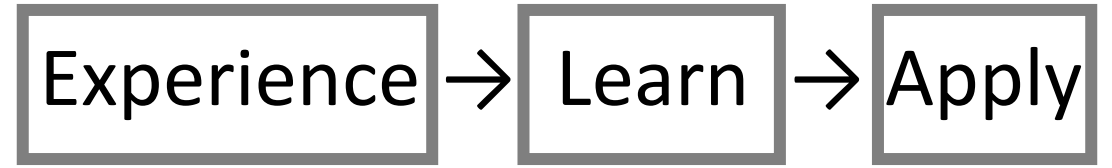


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What You'll Learn Today



What You'll Learn Today

Experience → Learn → Apply

- (35%) Experience how feedback shapes:
 - Attention, motivation, and engagement
 - Not just AI-mediated feedback

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- (35%) Learn what “AI-mediated feedback” means in media:
 - Difference from traditional feedback, why it is needed

What You'll Learn Today

Experience



Learn



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 - Responsible feedback design in media

What You'll Learn Today

Experience



Learn



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 - Not just AI-mediated feedback
- (35%) Learn what “AI-mediated feedback” means in media:
 - Difference from traditional feedback, why it is needed
- (30%) Apply insights to:
 - Responsible feedback design in media
- Bonus:
 - Related technologies, existing research

Quick Recall: Why Did You Click?

- Think about:
 - The **last notification** you clicked.
- What made it stand out?
 - Highlighted?
 - Urgent?
 - Personalized?



(Figure source: <https://support.apple.com>)

What Is Feedback?

Highlights

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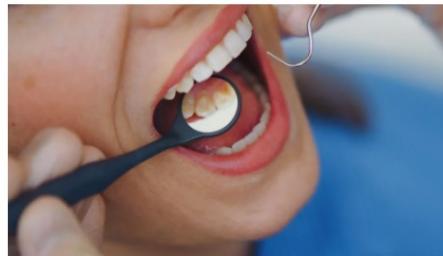
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Rankings



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What Is Feedback?

Update

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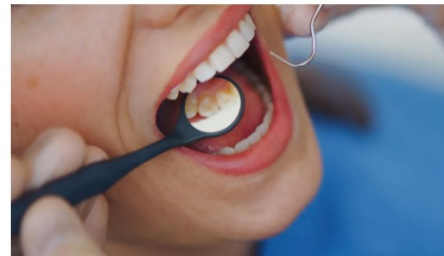
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What Is Feedback?

♥ Most Read/Love

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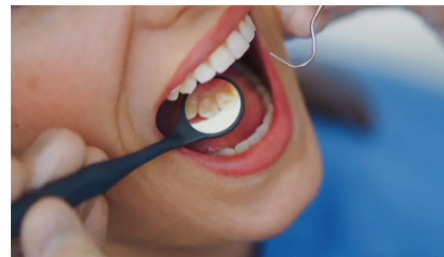
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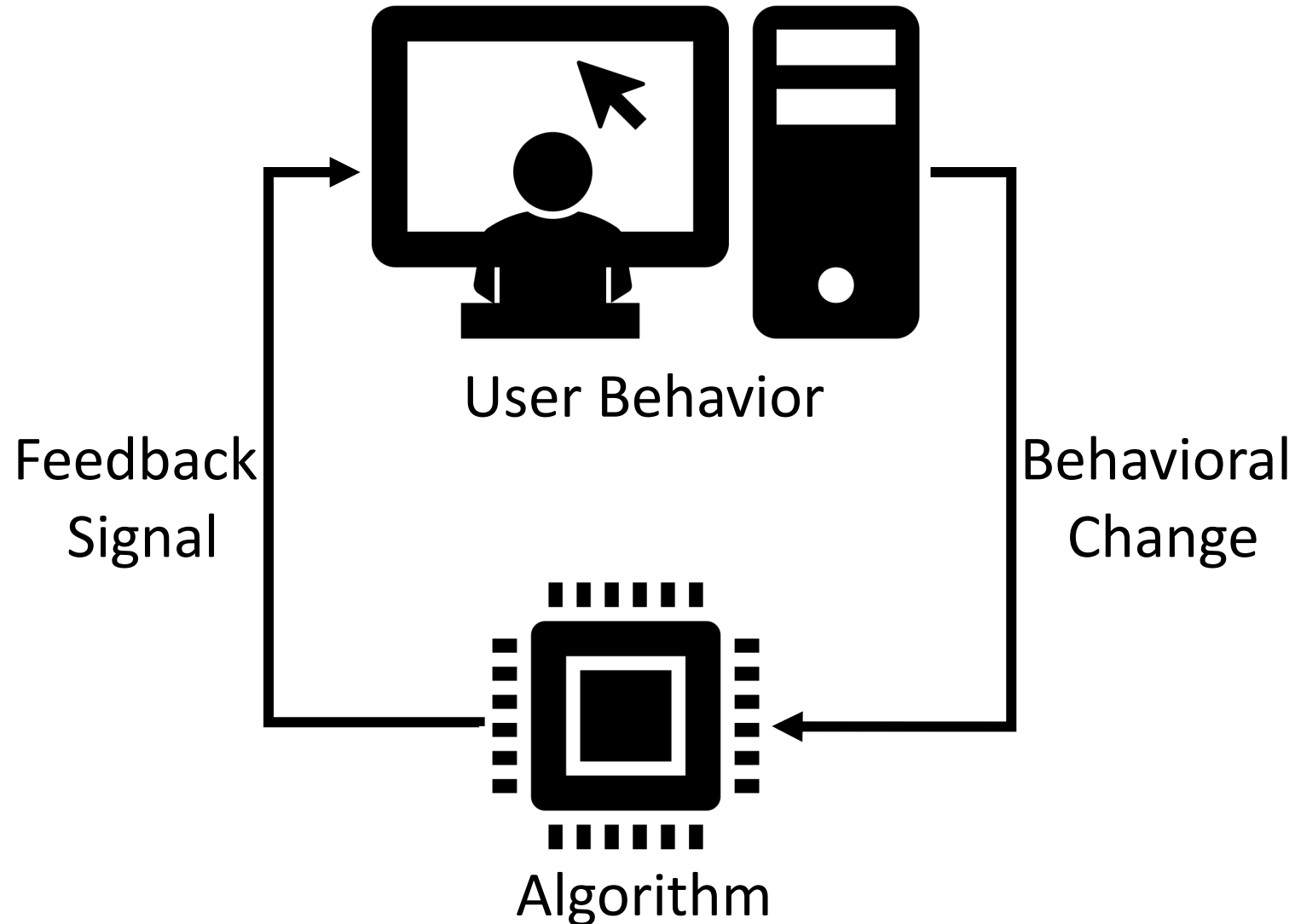
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What Is Feedback: Signals Regulating/Responding to User Behavior

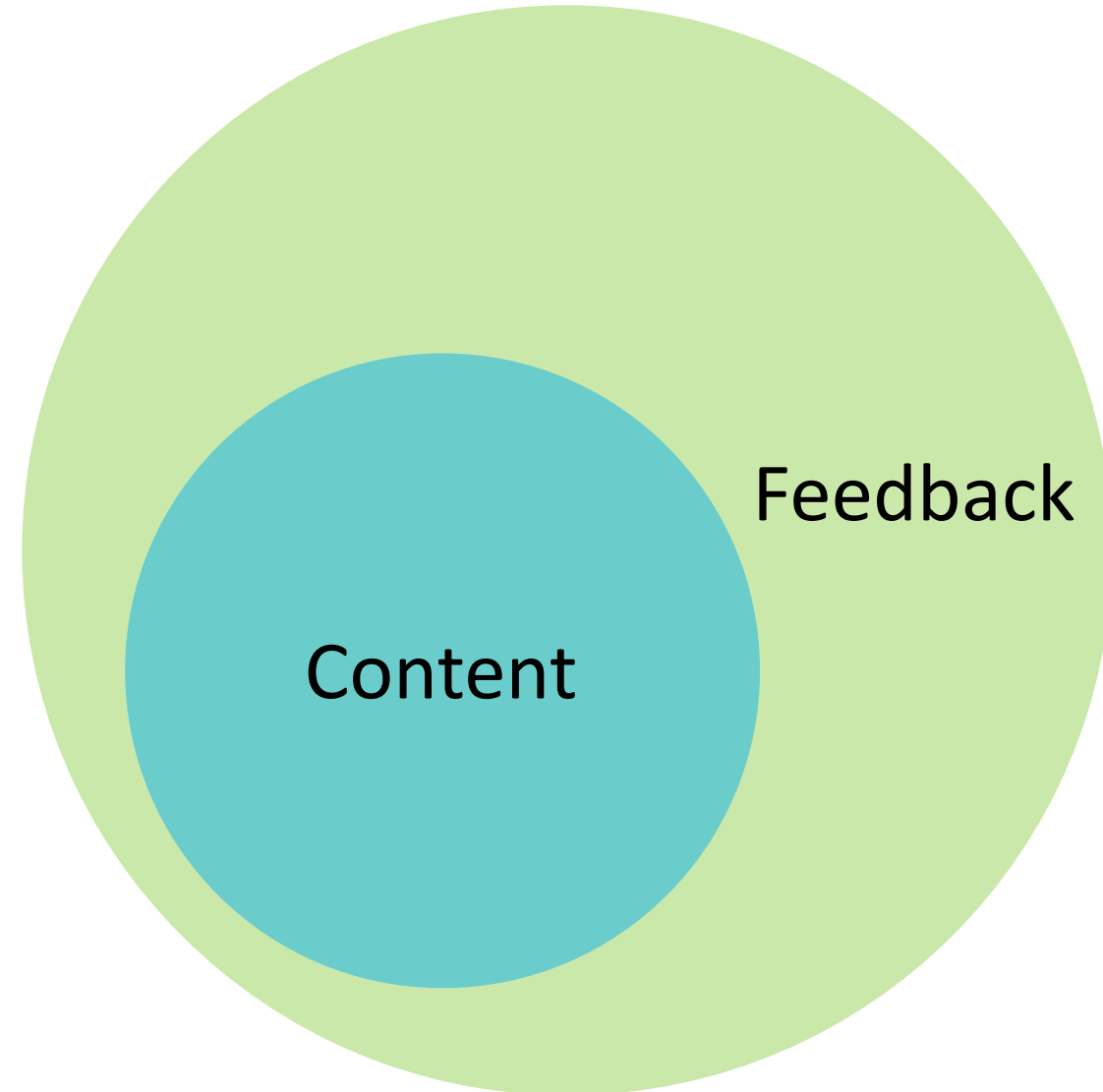
- Common examples:

- Rankings
("Top Stories")
- Highlights
- Notifications
- Engagement cues
("Most Read")



Content v.s. Feedback

Feedback contains content, but it is intended to **modulate behavior** rather than just communicate information.



Feedback Influences Your Behaviors

Feedback to Highlight



Fifteen cars have tyres slashed near school

The attack happened in a carpark in the Glenview Drive area of Limavady on Monday morning.



Traffic caused by £3.2m roadworks a 'big loss' for businesses

An Eglinton businesswoman has said the construction of a cycle path that "many feel will never be used" is costing her and other local businesses valuable customers.



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Feedback Influences Your Behaviors

No Feedback



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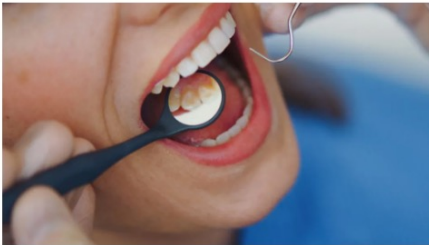
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Feedback Influences Your Behaviors

Feedback to Rank



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Activity: Reading Task in 30 Seconds

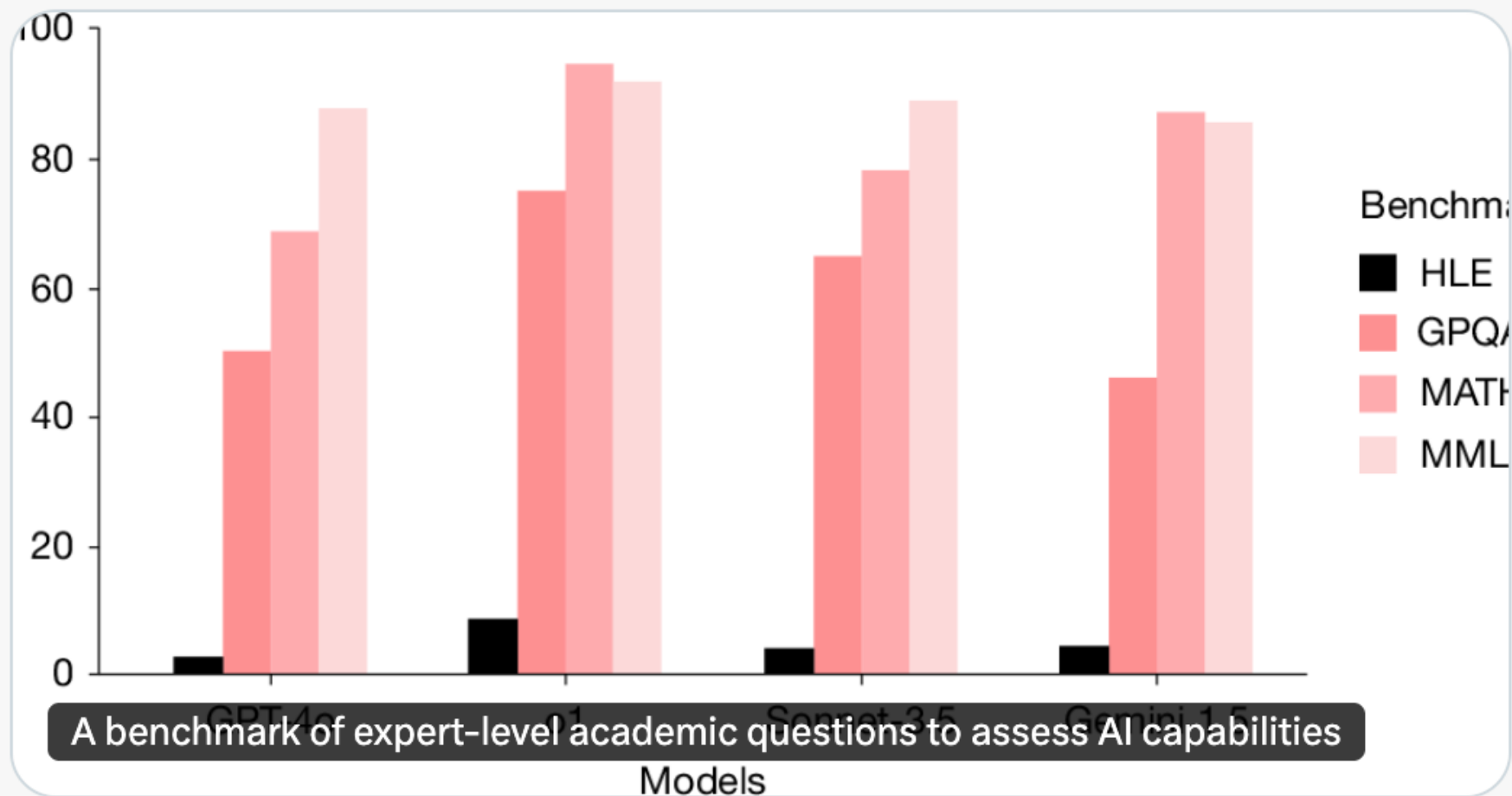
You'll read a short blog and then make quick decisions.



nature  @Nature · 7h



Nature research paper: A benchmark of **expert-level** academic questions to assess **AI** capabilities

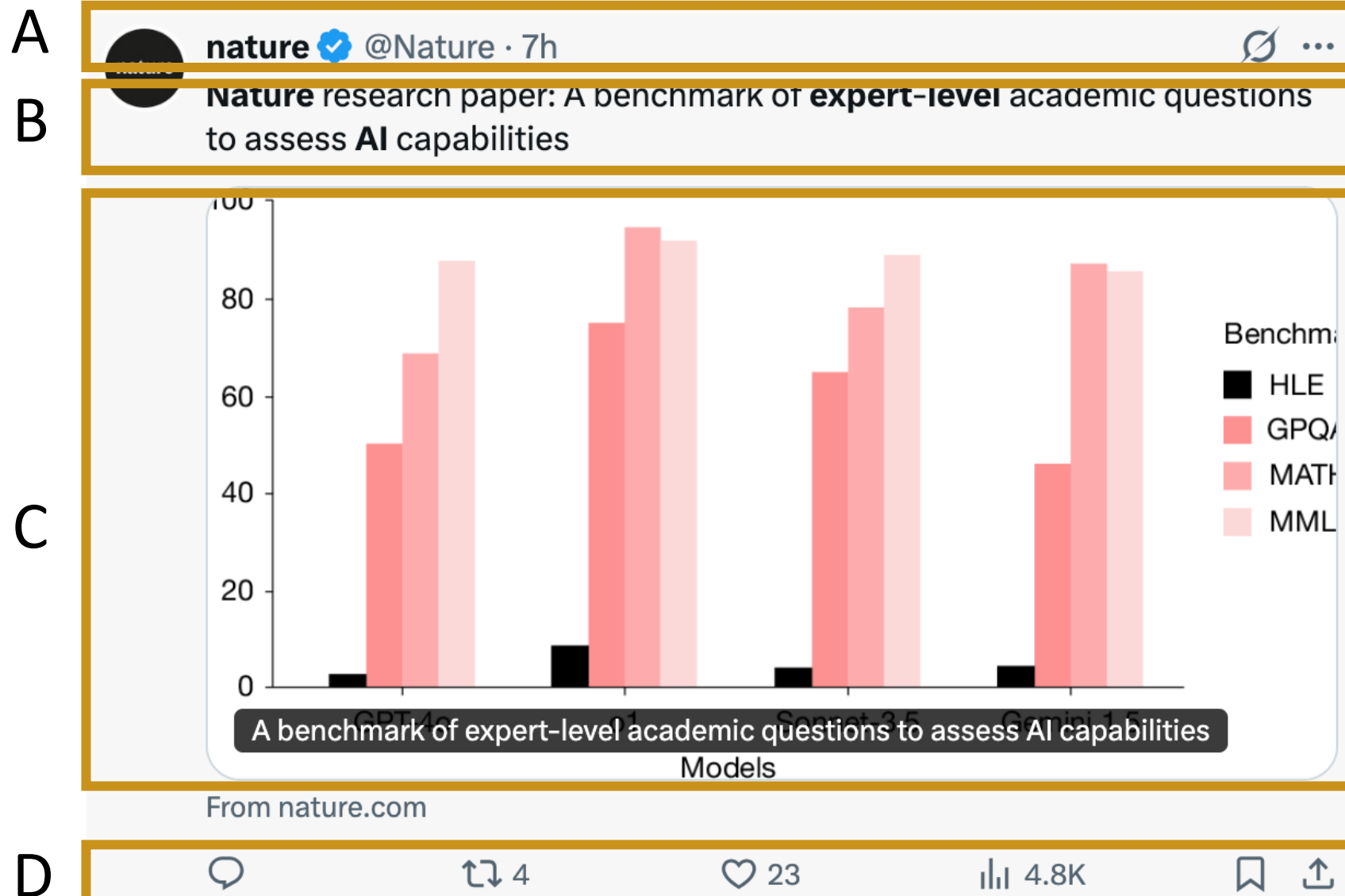


From nature.com



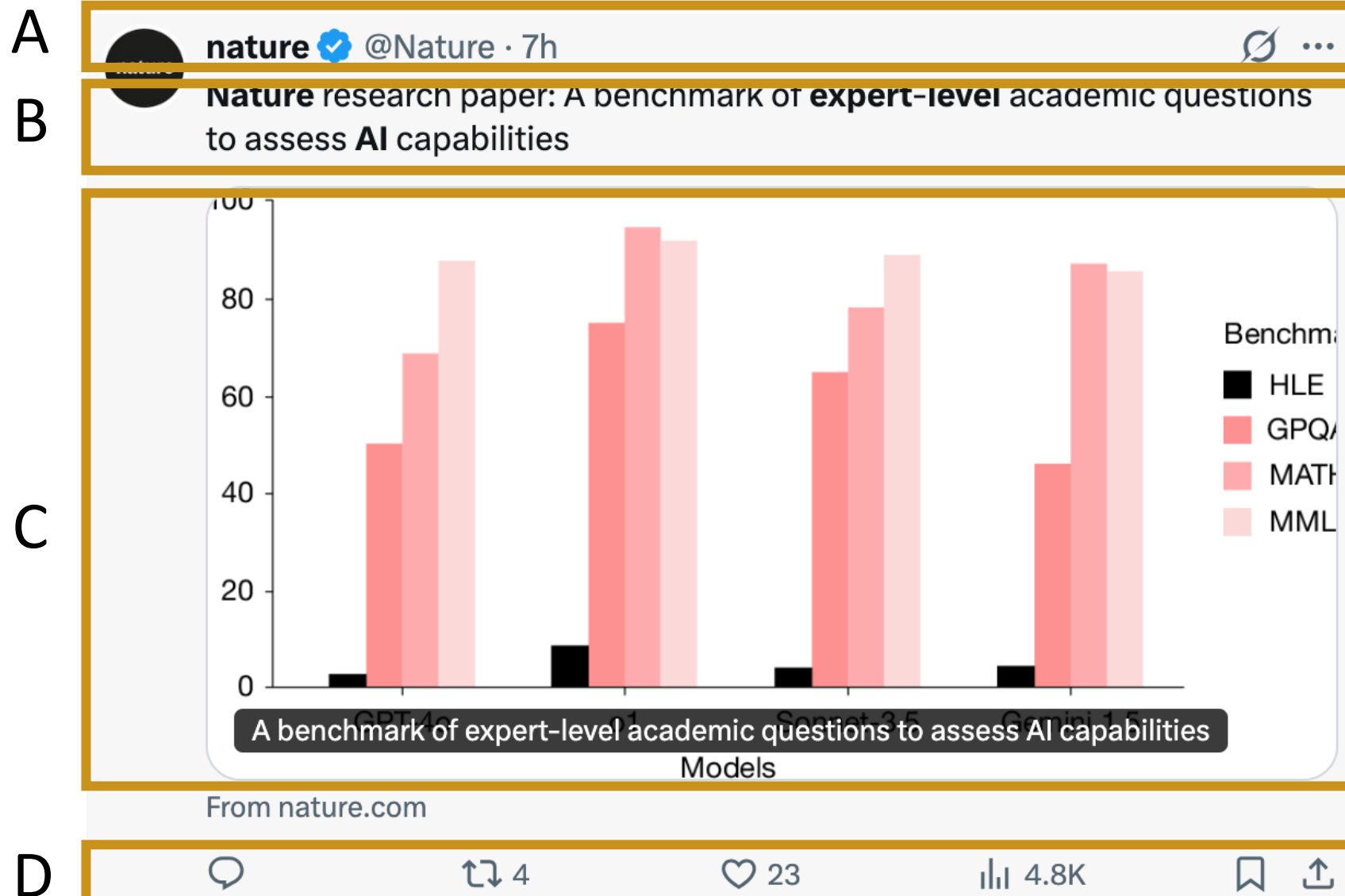
(Figure source: x.com)

Decision 1: “Which part did you read first?”



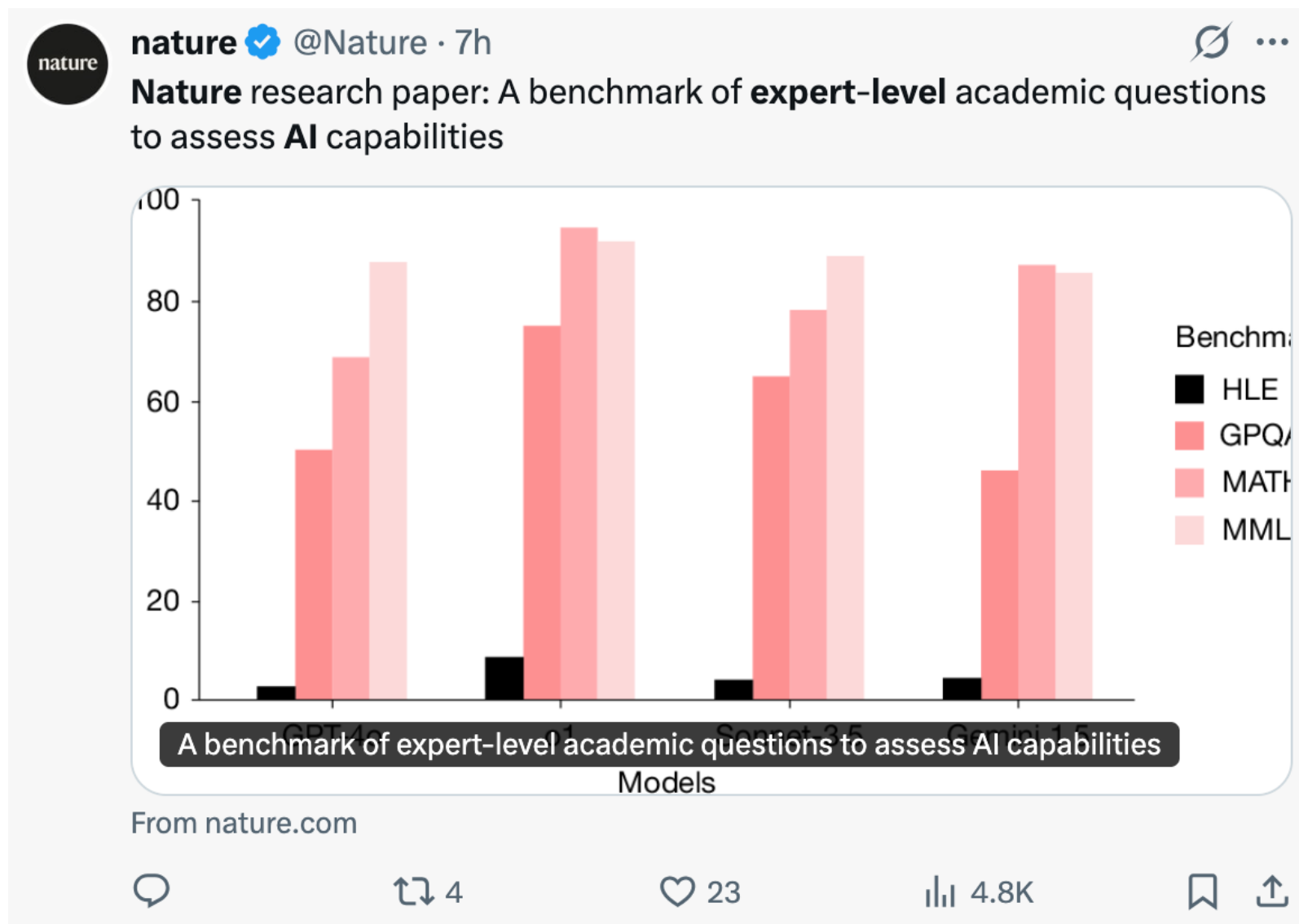
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Decision 1: Attention



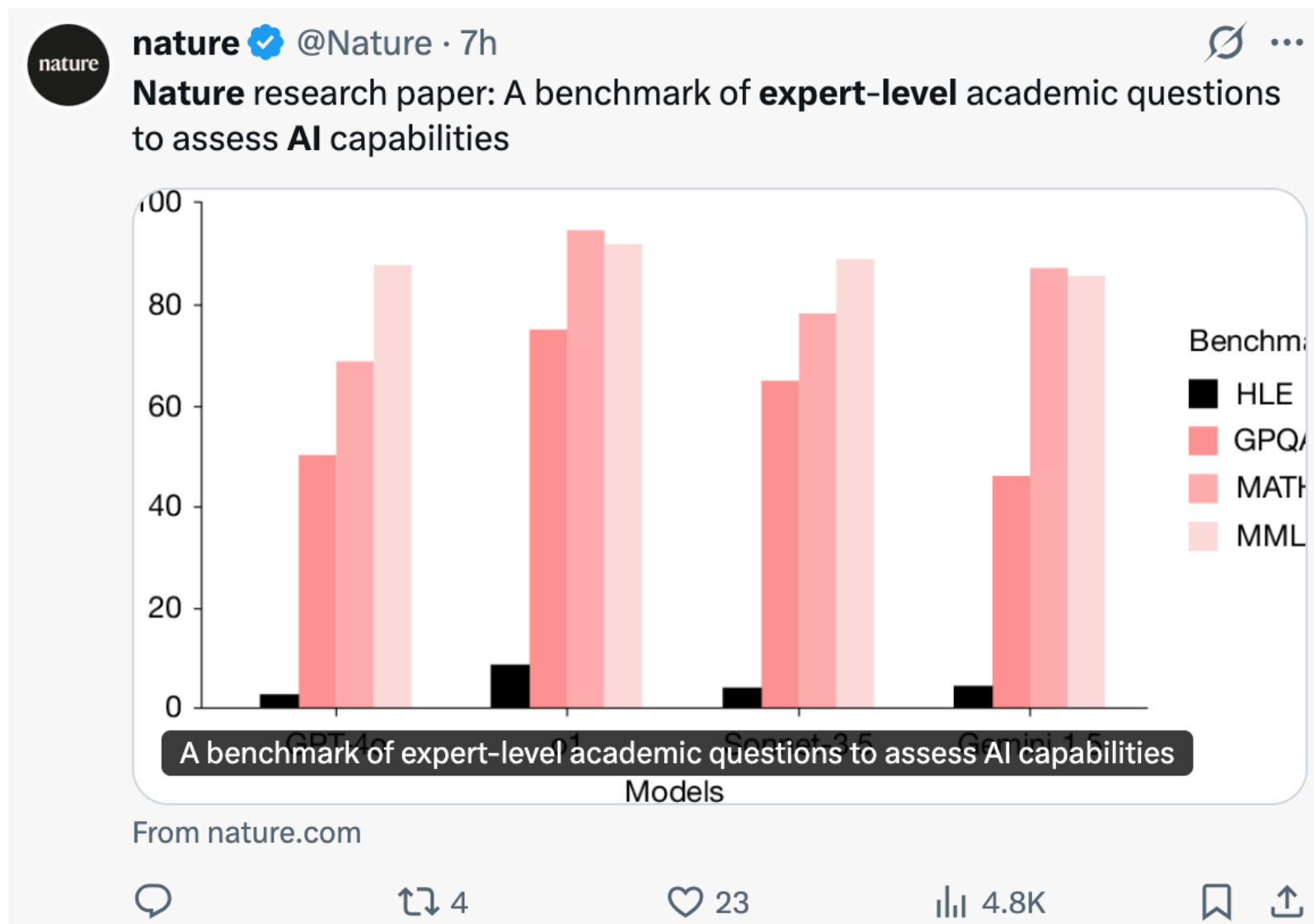
(Figure source: x.com)

Decision 2: “Will you click and keep reading?”



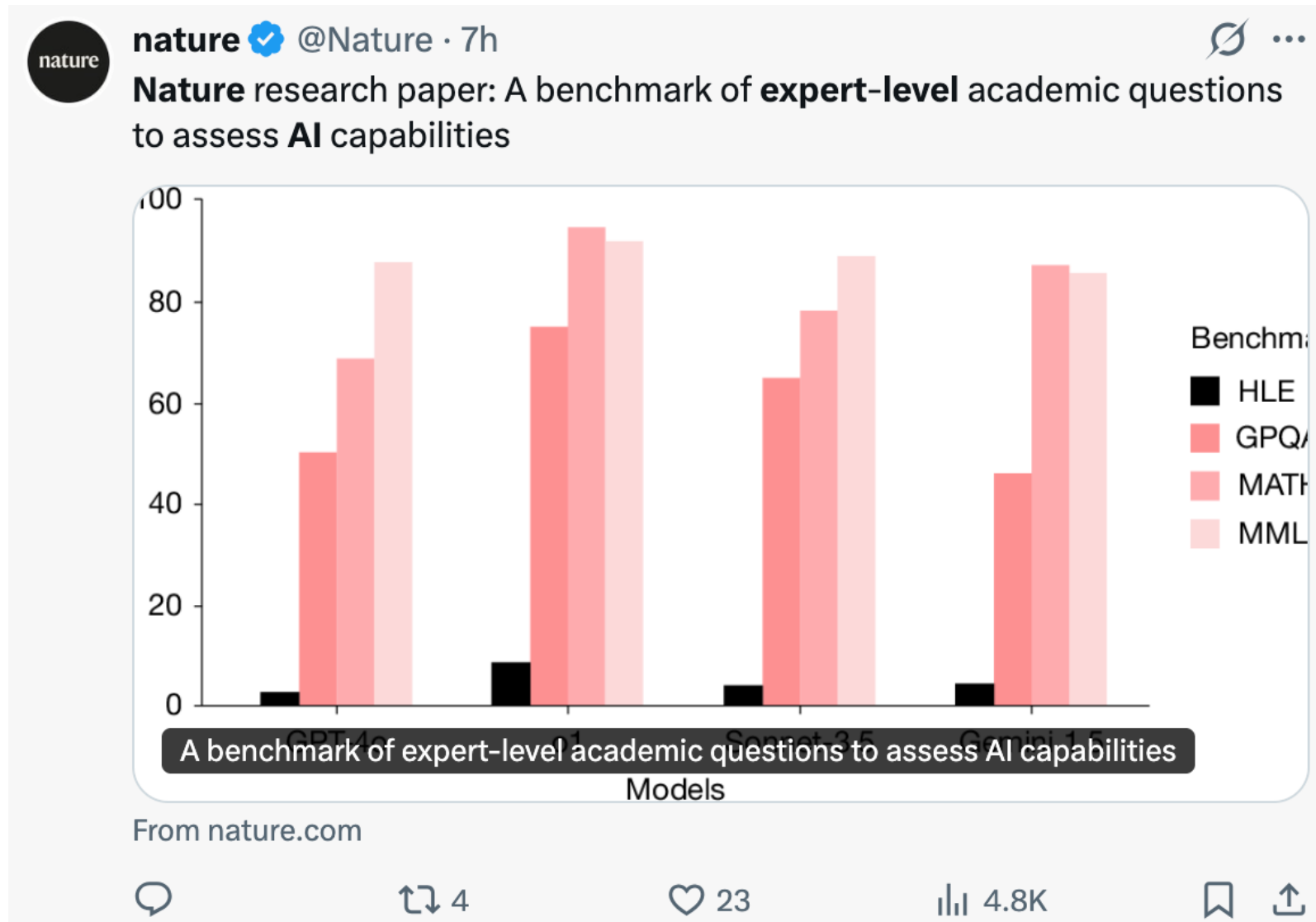
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Decision 2: Motivation



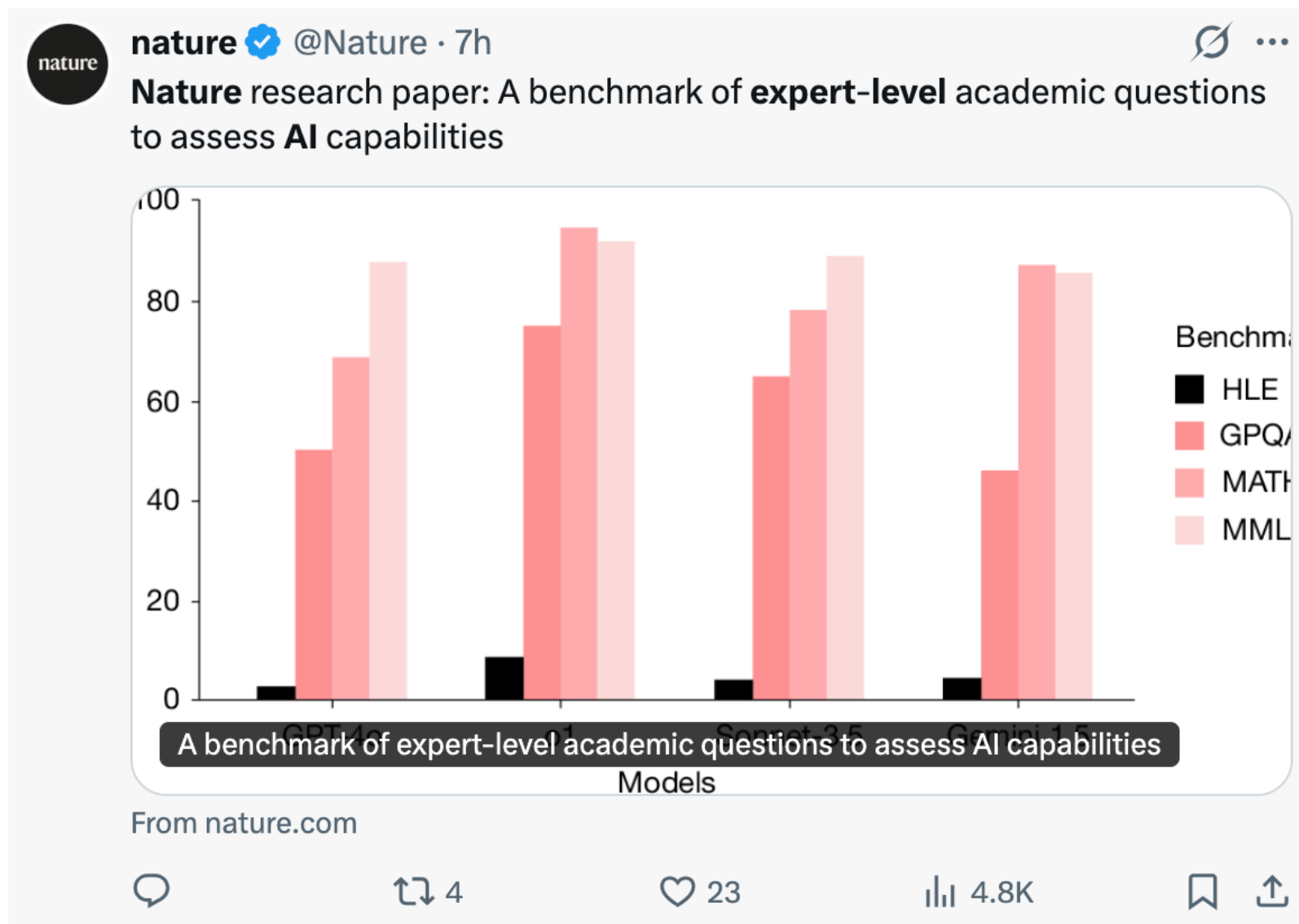
(Figure source: x.com)

Decision 3: “Would you comment on it?”



(Figure source: x.com)

Decision 3: Engagement



(Figure source: x.com)

Attention, Motivation, Engagement

- **Attention:** What you processed first
- **Motivation:** Whether you wanted to continue
- **Engagement:** Whether you would act

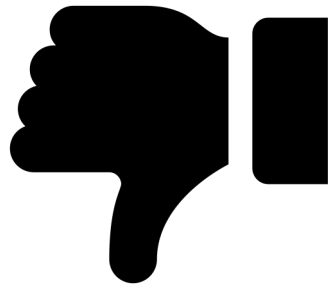
You Just Learned Through Feedback

- Experience differed across different feedback signals (textual/visual)
- Feedback influenced your behaviors (popularity, peer comments, highlights)

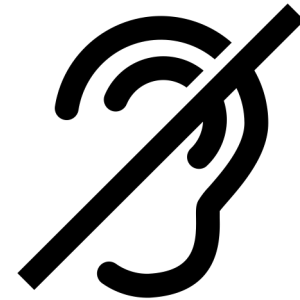
Diverse User Experience Under Feedback



Positive



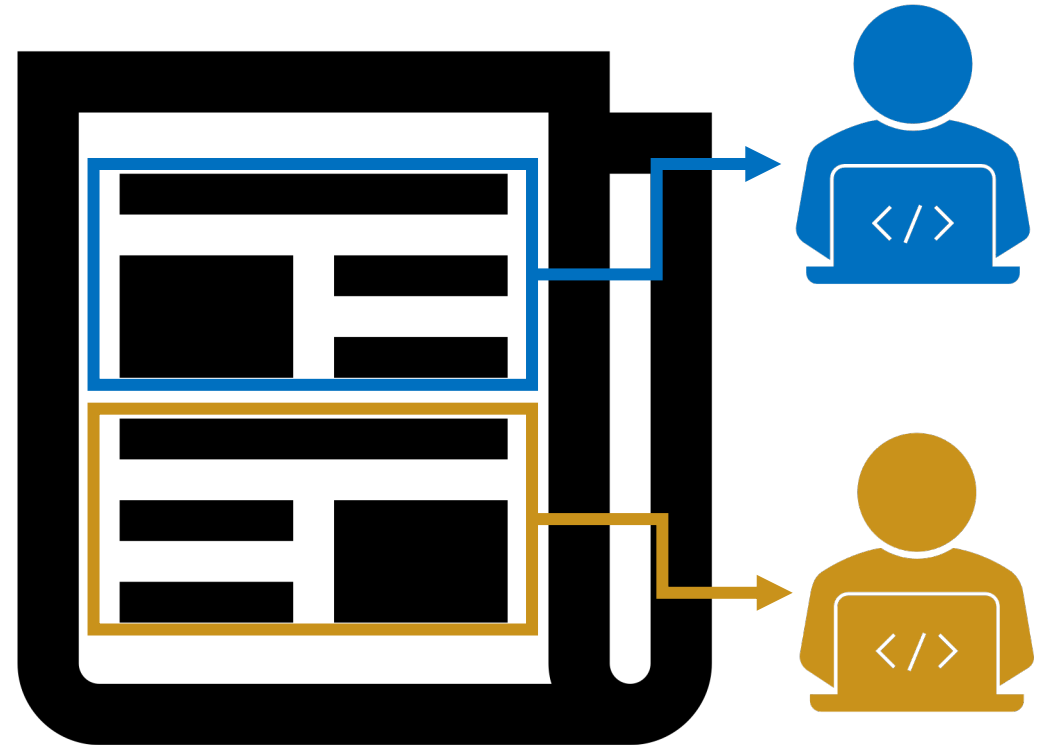
Negative



Neutral

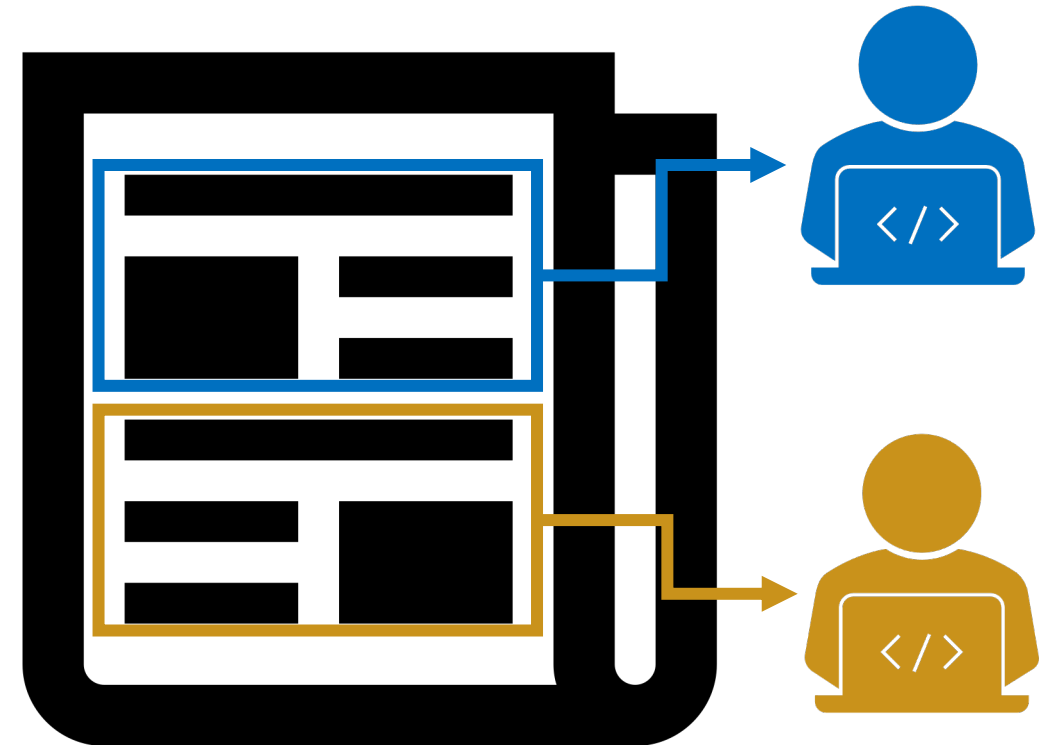
Why Do Platforms Need AI-Mediated Feedback?

- Users respond **differently** to the *same* feedback
- One-size-fits-all signals help some, harm others



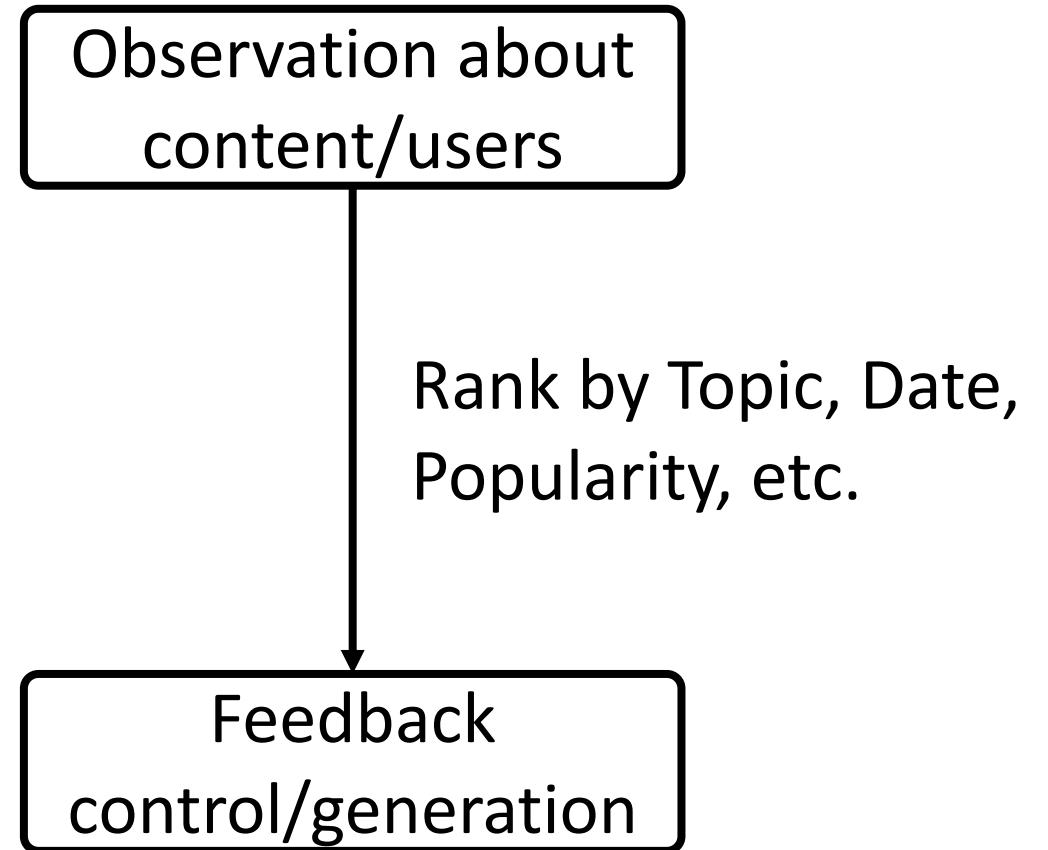
Why Do Platforms Need AI-Mediated Feedback?

- Users respond **differently** to the *same* feedback
- One-size-fits-all signals help some, harm others
- AI systems try to:
 - Predict individual attention
 - Adapt feedback dynamically
 - Maximize engagement *per user*



Rule-Based vs. AI-Mediated Feedback

- **Rule-based feedback**
 - Same signal for everyone
 - Predictable/Static

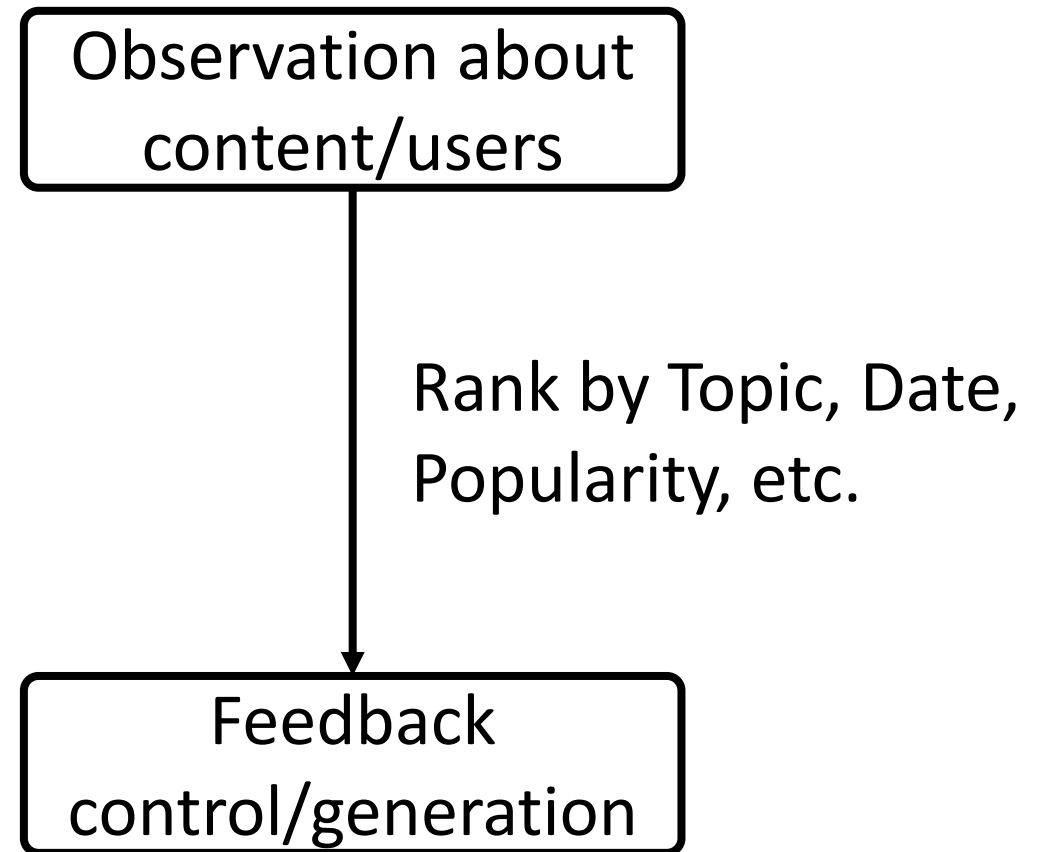


Rule-Based vs. AI-Mediated Feedback

- **Rule-based feedback**

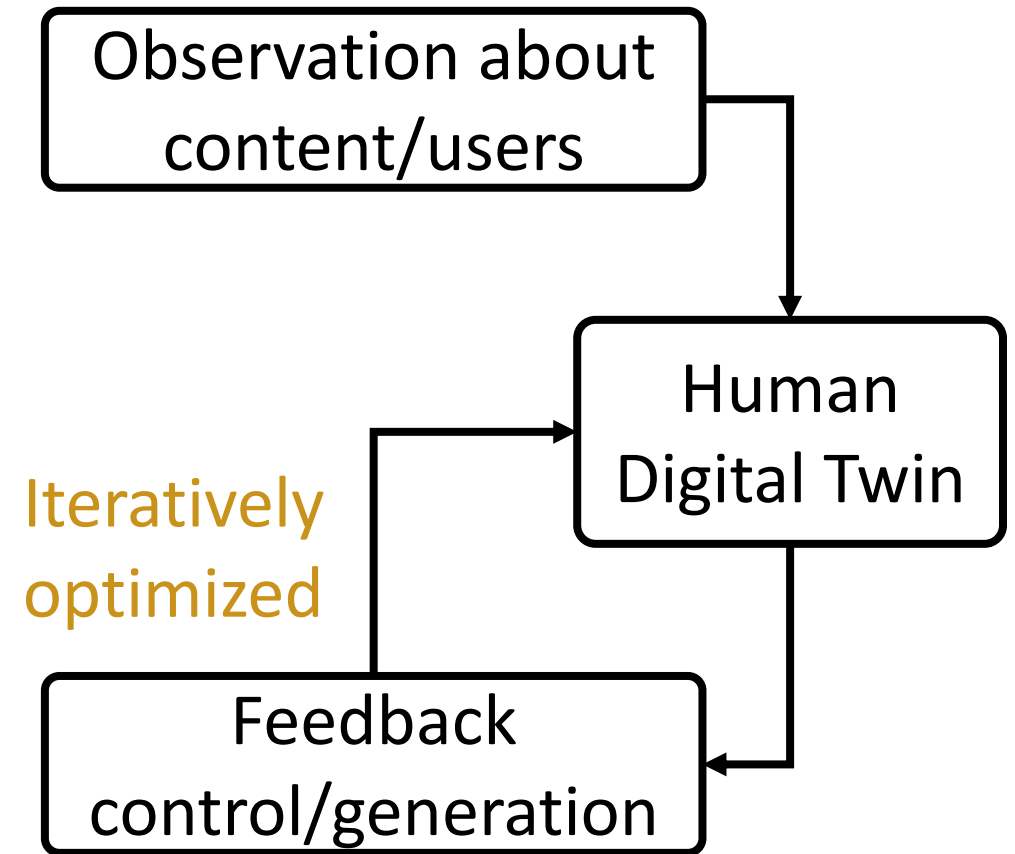
- Same signal for everyone
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Hand-coded rules;
Not adaptively optimized.

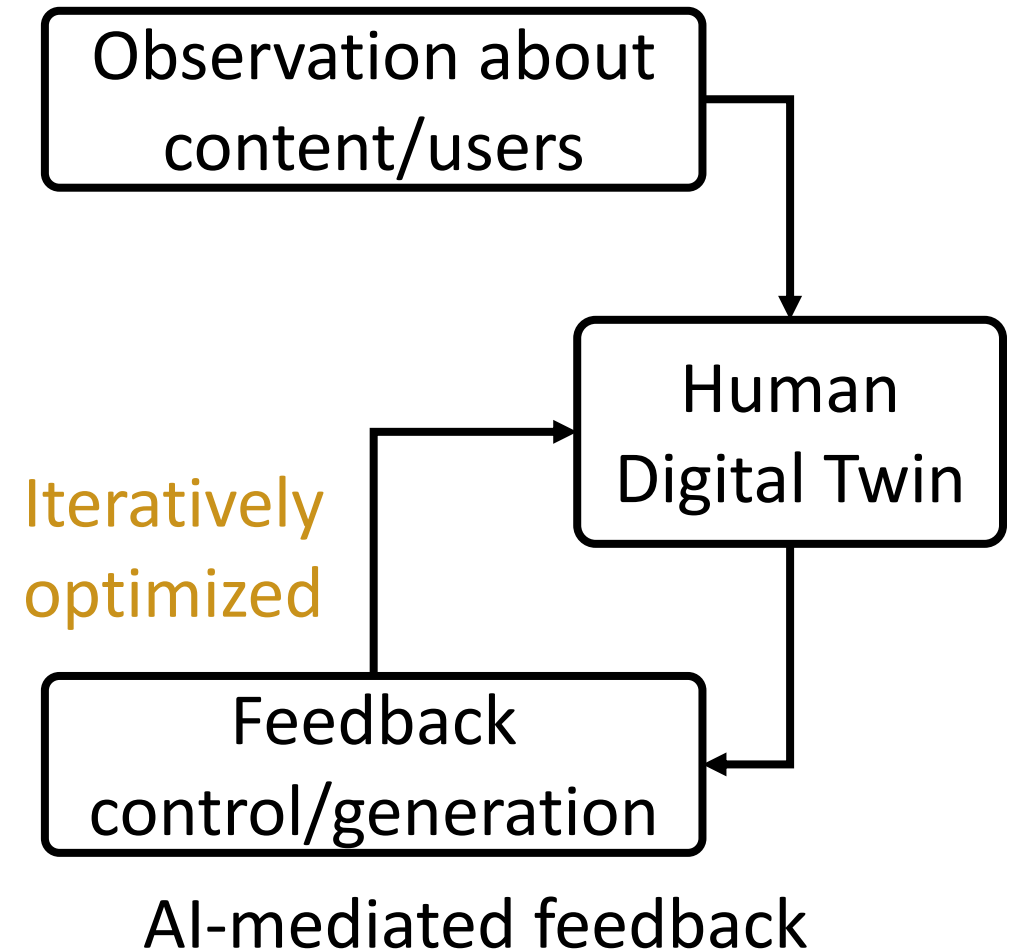
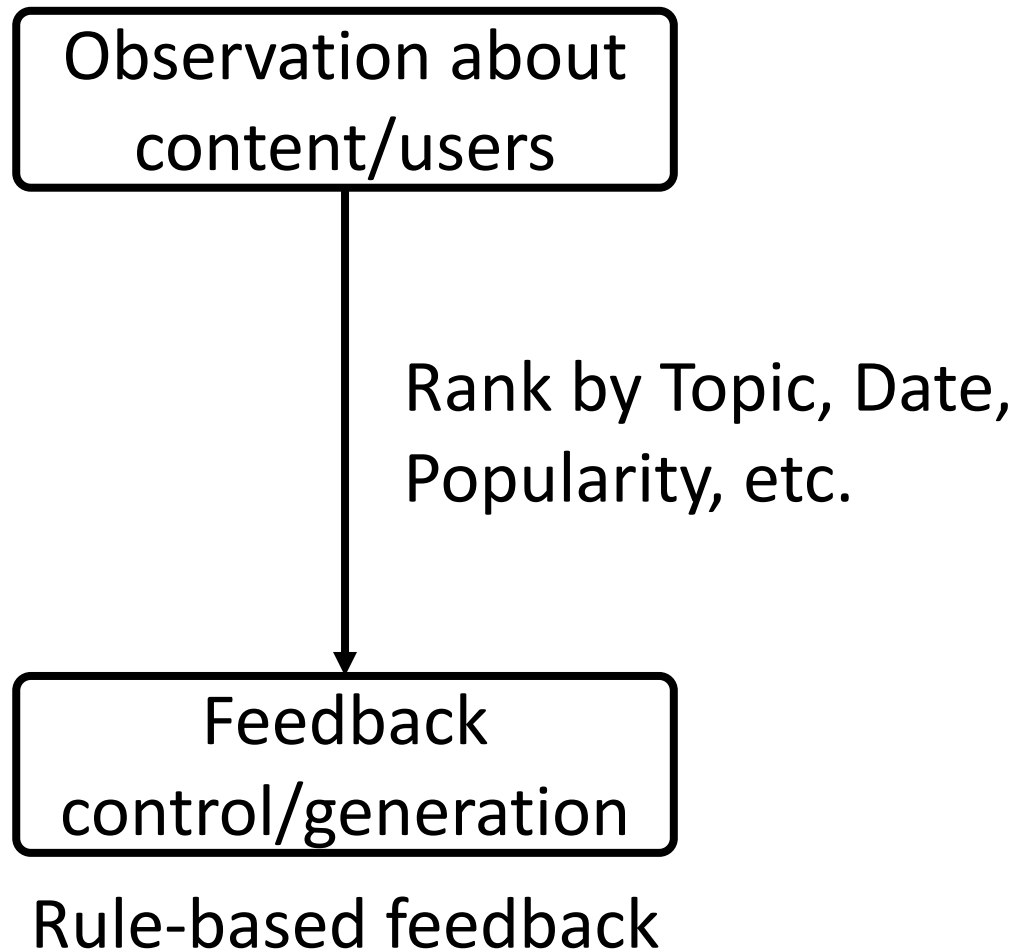


Rule-Based vs. AI-Mediated Feedback

- **AI-mediated feedback**
 - Learns from user behavior
 - Personalized and adaptive



AI-mediated feedback is learned from patterns, not hand-coded rules.



AI Feedback in Real Media Systems

- “Breaking news” alerts

Today's News ×

Tesla to End Model S and X Production for Optimus Robots



9 hours ago · News · 47.2K posts

Moonshot AI Launches Kimi K2.5 as Top Open-Source Model



2 days ago · Other · 29.7K posts

Fed Keeps Rates Steady at 3.50%-3.75% as Inflation Persists



1 day ago · News · 74.5K posts

(Figure source: x.com)

AI Feedback in Real Media Systems

- Trending topics

What's happening

Galaxy Z Fold7

...

Big screen. Big AI edits.

🔗 Promoted by Samsung Mobile US

Trending in United States

...

Crescent

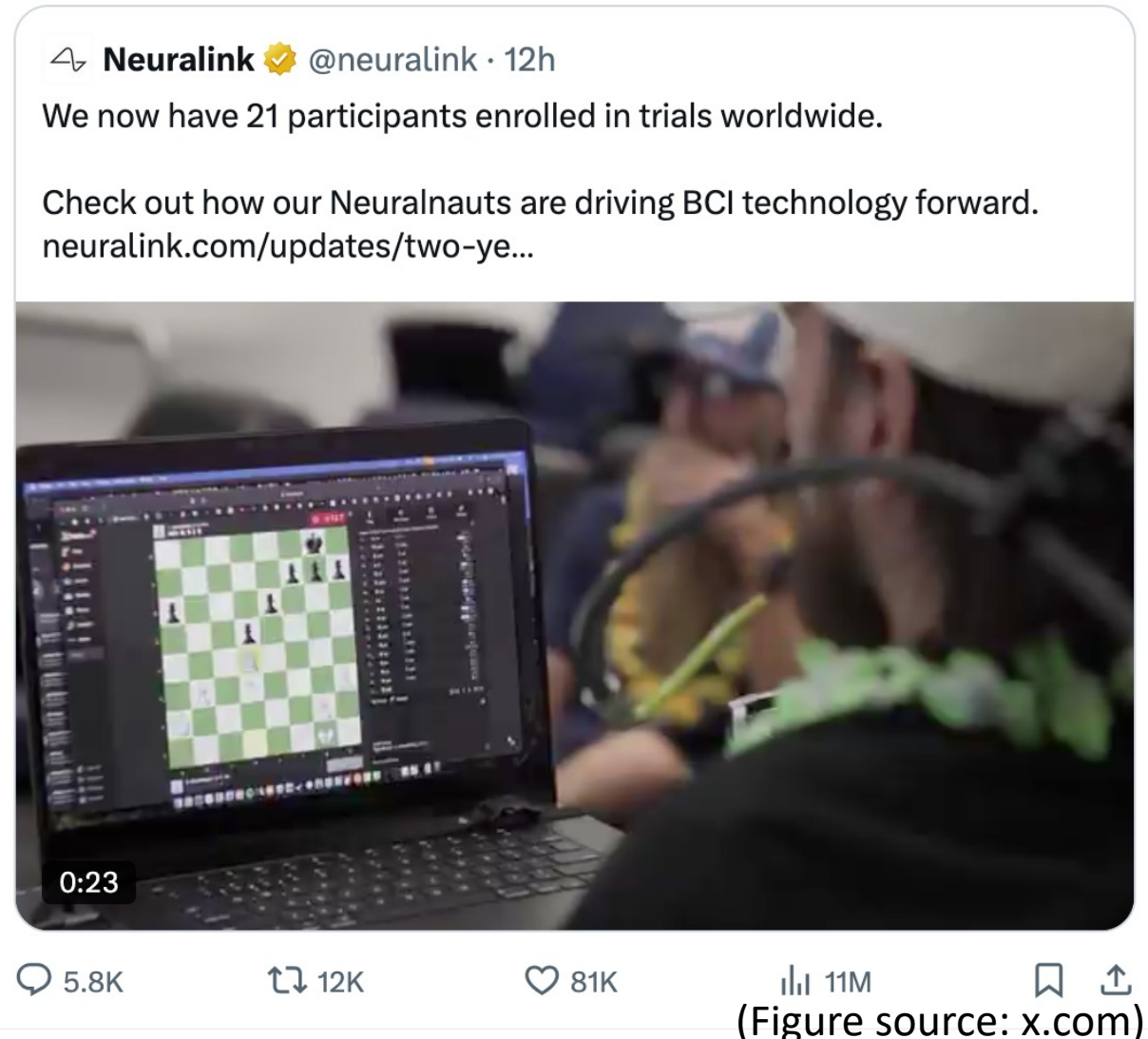
Technology · Trending

...

Windows 11

AI Feedback in Real Media Systems

- Engagement metrics for:
 - journalists
 - users



AI Feedback in Real Media Systems

- Ranked comments or headlines



(Figure source: x.com)

AI Feedback in Real Media Systems

- “Breaking news” alerts
- Trending topics
- Engagement metrics for:
 - journalists
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- Ranked comments or headlines
- ...

Designing Feedback Responsibly

- Feedback shapes public attention and makes impact

 5.8K

 12K

 81K

 11M

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 5.8K

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- Public attention $\neq$ public interest

Designing Feedback Responsibly

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💬 5.8K

↻ 12K

❤️ 81K

📊 11M

- Public attention ≠ public interest
- Key question:
 - What feedback should we redesign—or remove?

Engagement

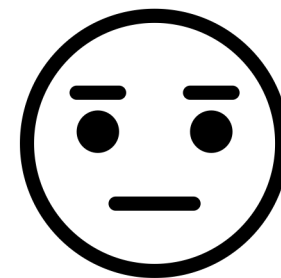
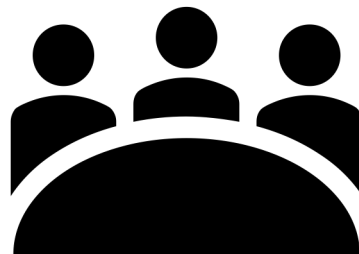
Responsibility



Balance

Ethical Questions: Who Should Shape Attention?

- AI feedback can:
 - **Amplify certain voices** while silencing others
 - Optimize engagement, not necessarily truth or public interest
 - **Shape emotions**: urgency, fear, confidence, or stress



Ethical Feedback Design: Preserve User Autonomy

- Avoid false urgency
("Breaking" inflation)
 - Manipulating emotions rather than informing.
 - Alert fatigue.

Ethical Feedback Design: Preserve User Autonomy



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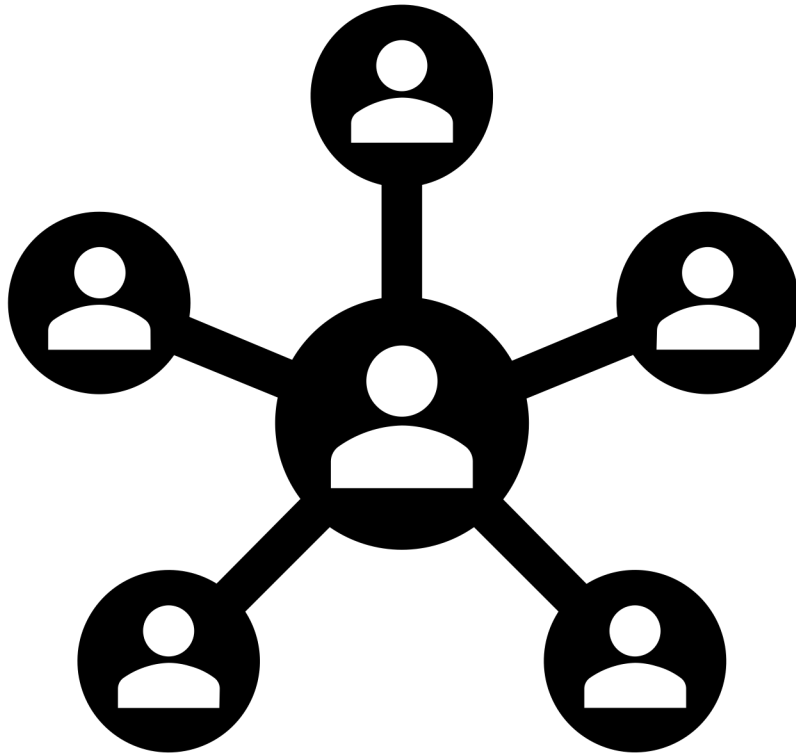
"If everything is breaking, nothing is."

Ethical Feedback Design: Support Diverse Attention

- Prevent runaway amplification
 - Collapses attention into a single emotion.
 - Creates informational monocultures.
 - Reinforces extreme worldviews.

Ethical Feedback Design: Support Diverse Attention

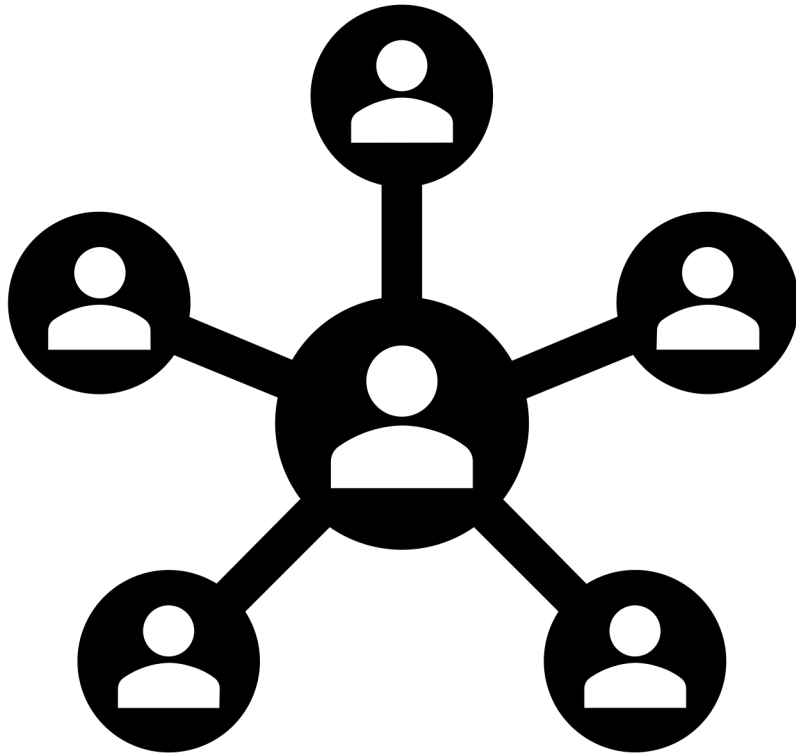
Algorithmic boosting of
a single viral narrative



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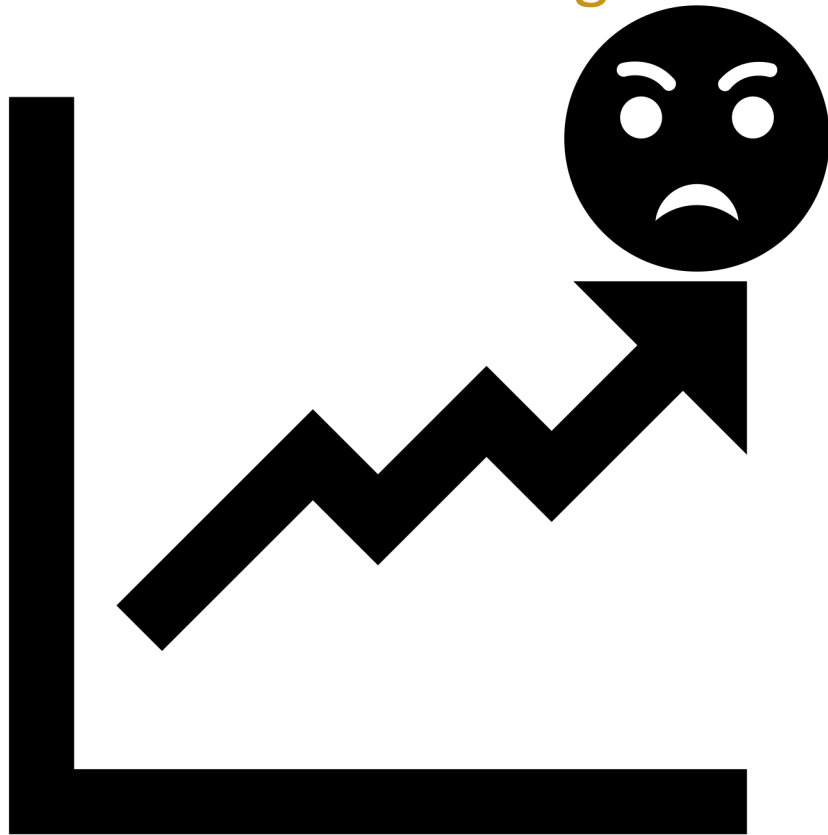
“Attention diversity is a public good—but algorithms treat it like noise.”

Ethical Feedback Design: Design for well-being

- Avoid exploiting outrage
 - Increases anxiety, and burnout
 - Social interaction →
emotional extraction

Ethical Feedback Design: Design for well-being

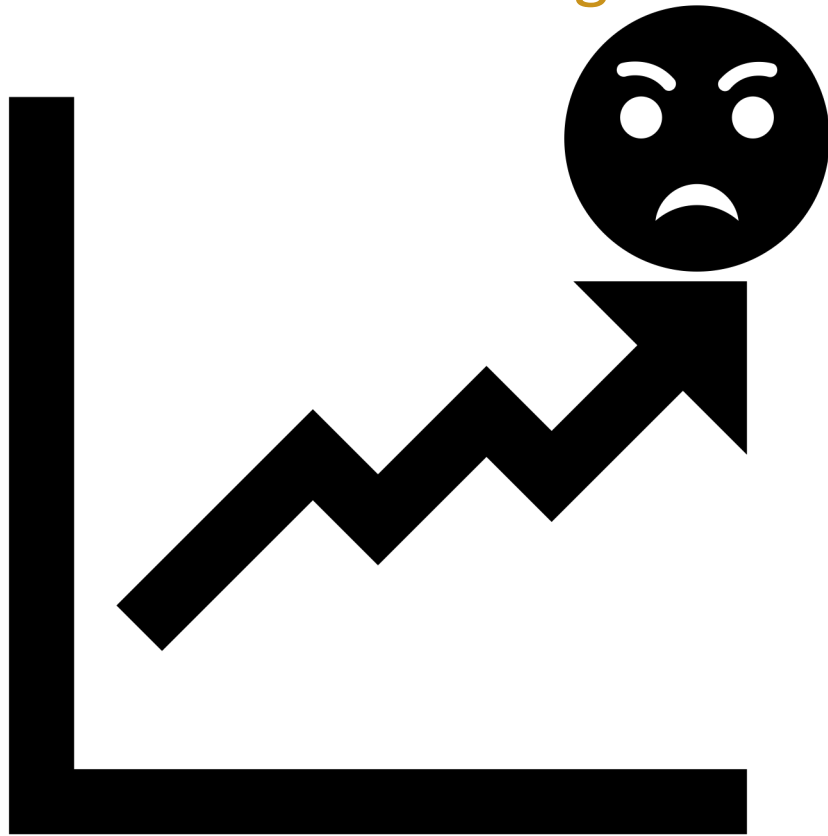
Feedback algorithm that
rewards outrage



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Ethical Feedback Design: Design for well-being

Feedback algorithm that
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“What we reward, we reproduce.”

Ethical Feedback Design



- Preserve user autonomy
 - Avoid false urgency (“Breaking” inflation)



- Support diverse attention
 - Prevent runaway amplification



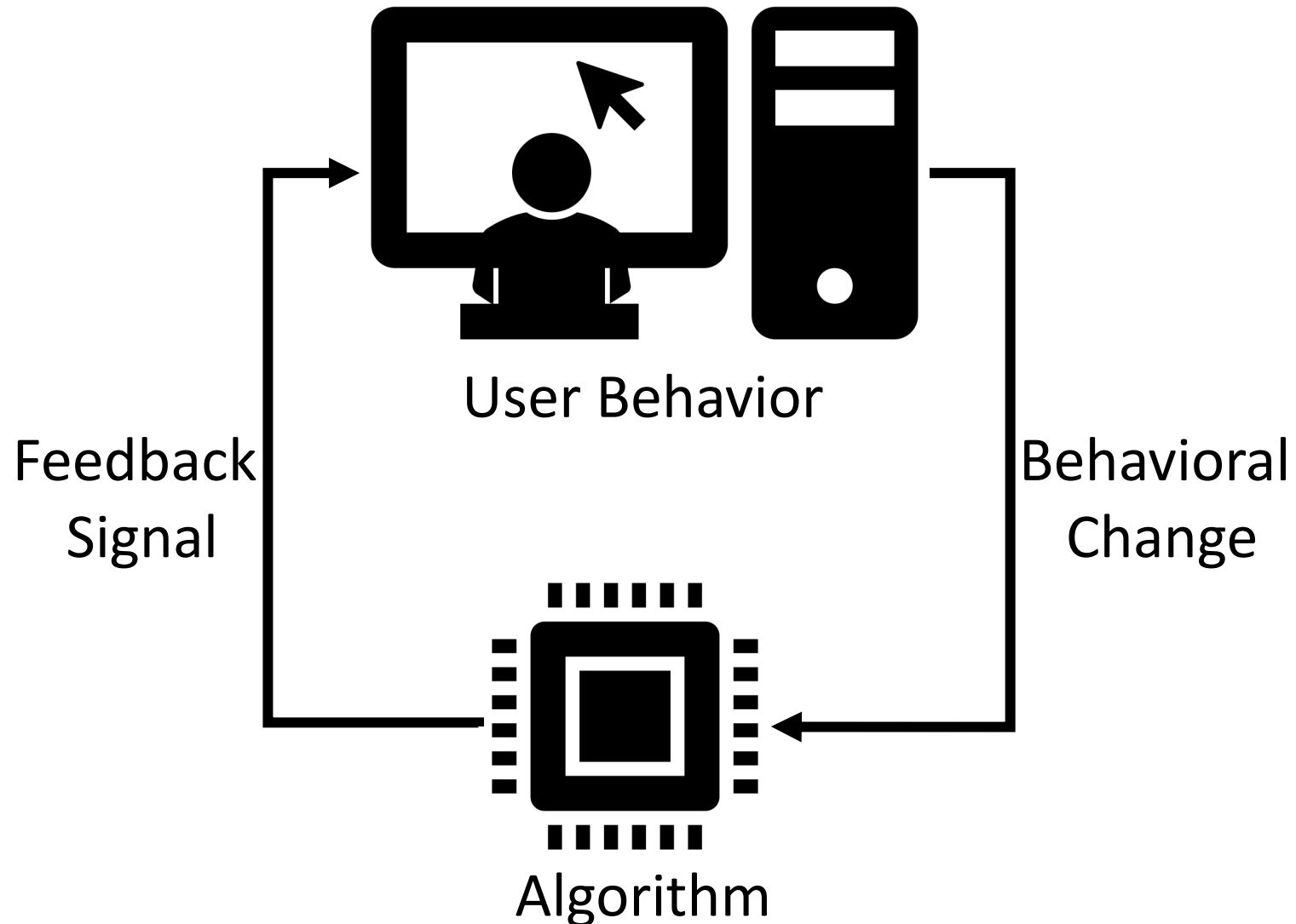
- Design for well-being
 - Avoid exploiting fear or outrage

Ethical Feedback Design

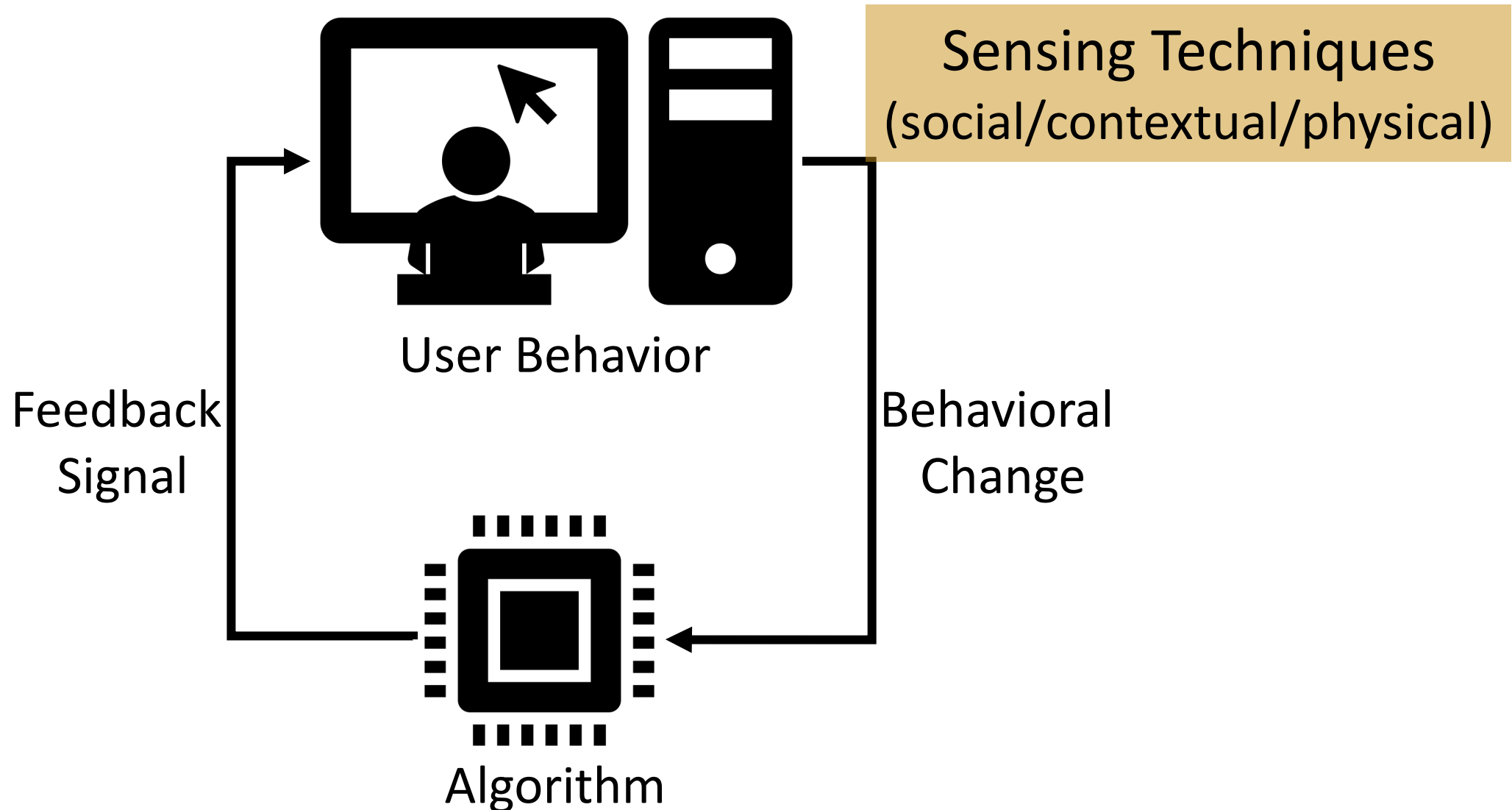
Feedback is ethical during optimization when it:

- ✓ Empowers informed attention
- ✓ Respects individual differences
- ✓ Serves collective well-being

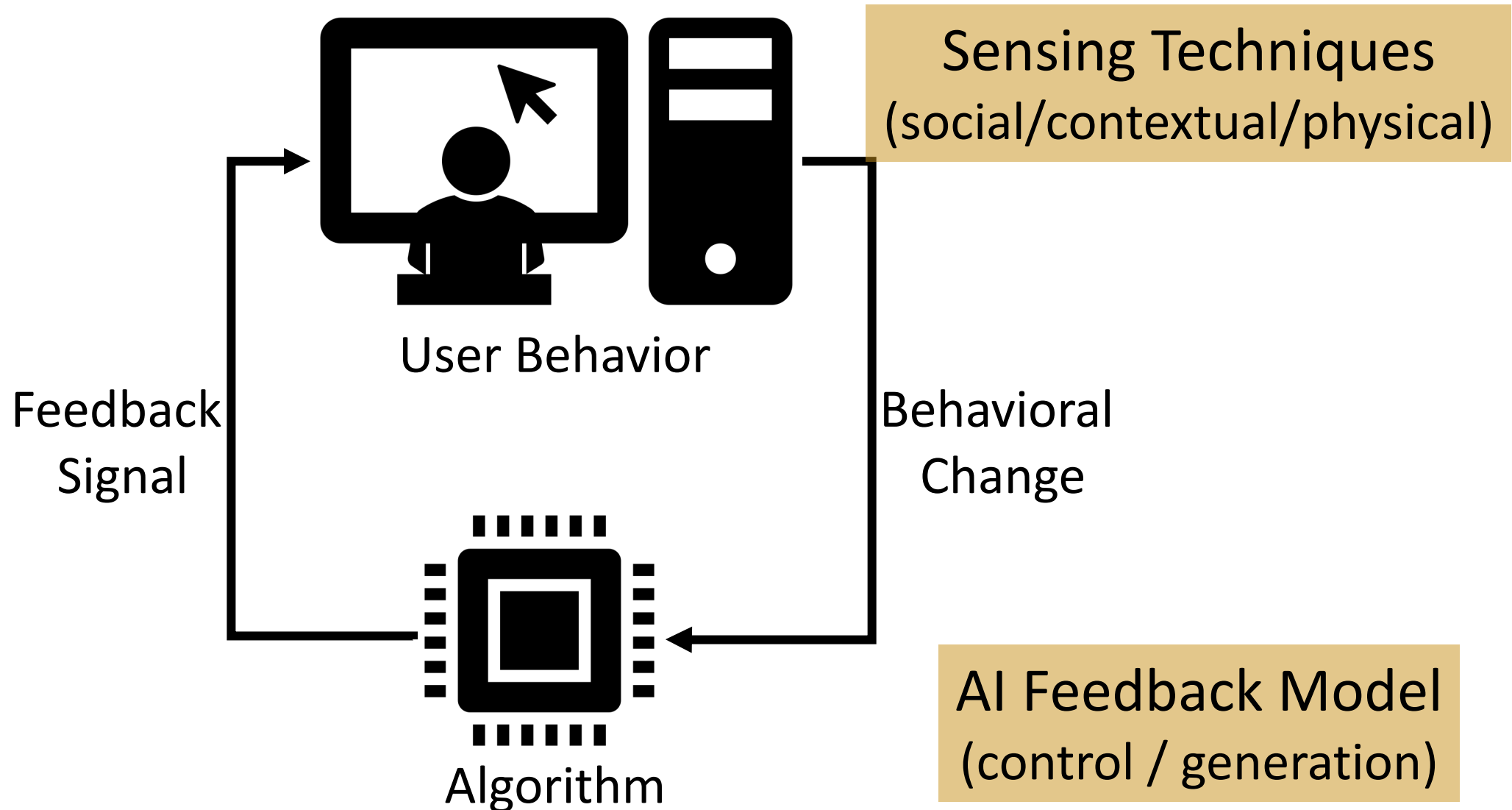
Bonus: What Technologies Power AI-Mediated Feedback?



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Bonus: Existing Research Efforts: Behavioral Change

PNAS

ARTICLES ▼

FRONT MATTER

AUTHORS ▼

TOPICS +

RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES | 

Perceived social presence reduces fact-checking

Jun et al. 2017

Bonus: Existing Research Efforts: Useful Sensing Signals

nature human behaviour



Article

<https://doi.org/10.1038/s41562-024-02065-6>

Web-browsing patterns reflect and shape mood and mental health

Kelly et al. 2024

Bonus: Existing Research Efforts: AI to Deliver Peer Feedback

nature machine intelligence

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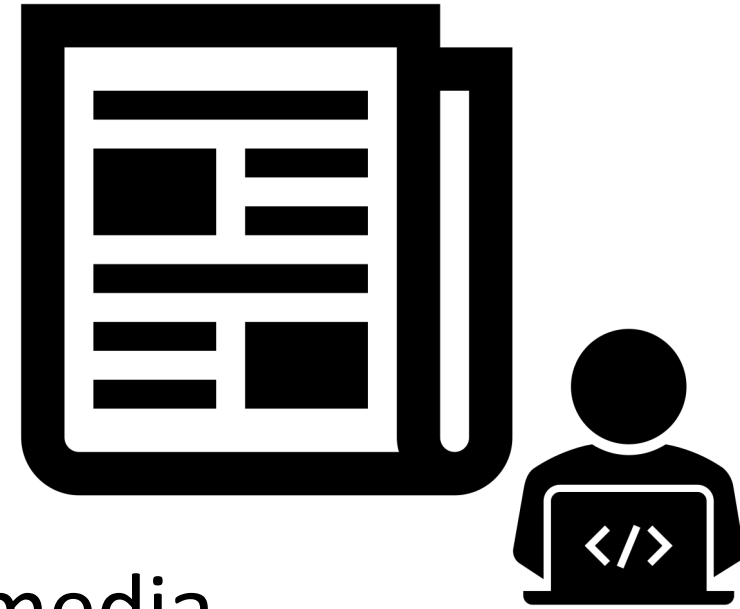
Human–AI collaboration enables more empathic conversations in text-based peer-to-peer mental health support

Sharma et al. 2023

Bonus: Existing Research Efforts

Feel free to check more interesting papers
after this lecture!

Takeaway Message



- Feedback shapes:
 - Attention, motivation, and engagement
- “AI-mediated feedback” is important in media
- Responsible feedback design in media
- Bonus
 - Related technologies, existing research